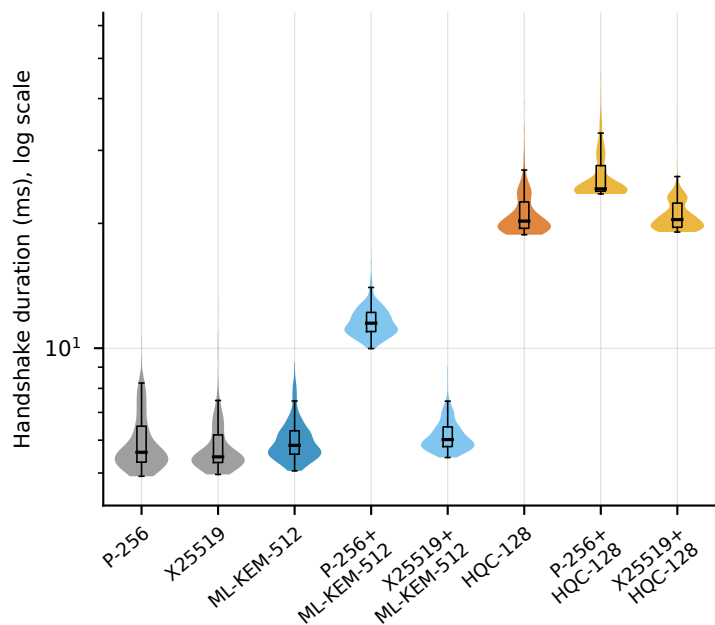
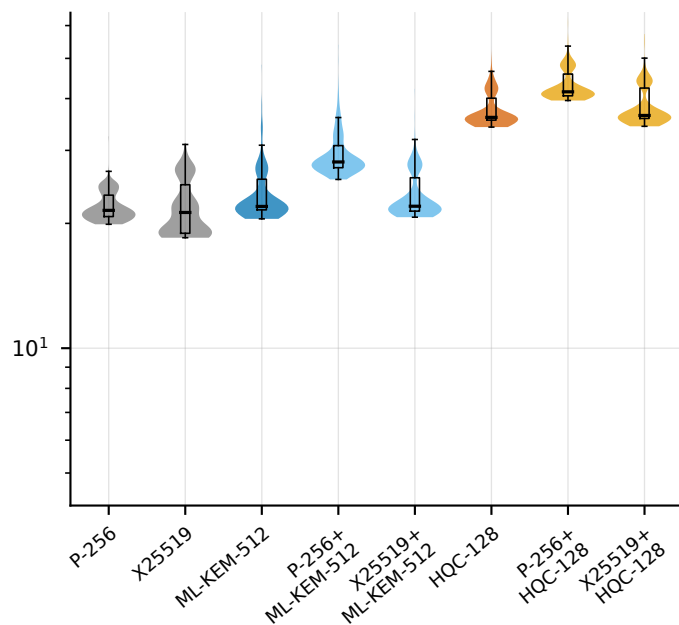


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	6.03	16.7	10.6	22.0	7.7	32.4
X25519	5.84	15.9	13.1	21.9	15.4	31.0
ML-KEM-512	6.05	12.0	9.27	23.6	15.2	48.3
P-256+ML-KEM-512	11.7	8.5	17.4	29.5	12.3	53.9
X25519+ML-KEM-512	6.19	9.0	9.04	23.5	12.9	42.2
HQC-128	21.2	12.1	39.3	37.5	9.9	70.3
P-256+HQC-128	26.1	14.4	46.4	43.2	9.3	80.2
X25519+HQC-128	21.0	9.5	37.9	38.3	10.7	57.3